



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

The data is collected through SpaceX Api and Web Scraping. Data Wrangling and Exploratory Analysis are performed on the collected data by using appropriate python modules and SQL. The data visualizations are used to further understand data. Both EDA and Visual Analytics using Folium and Interactive dashboards with Plotly Dash are use. At last different ML Algorithms are investigated to decide best model for future predictions.

EDA on collected indicate increased success rate over time. Success rate also depends on launch site and orbit type.

Introduction

The next big thing in science and technology is space exploration. One of the leading company in this field is SpaceX. They claim the cost of Falcon 9 rocket to be 62 million dollars compared to 165-million-dollar cost for the rivals.

The objective of this research is to investigate the success probability of landing of the first stage. For our company SpaceY to compete with SpaceX, we need to perform better than SpaceX. Therefore, determining success rate and hence cost for SpaceX is critical.

Section 1

Methodology

Methodology

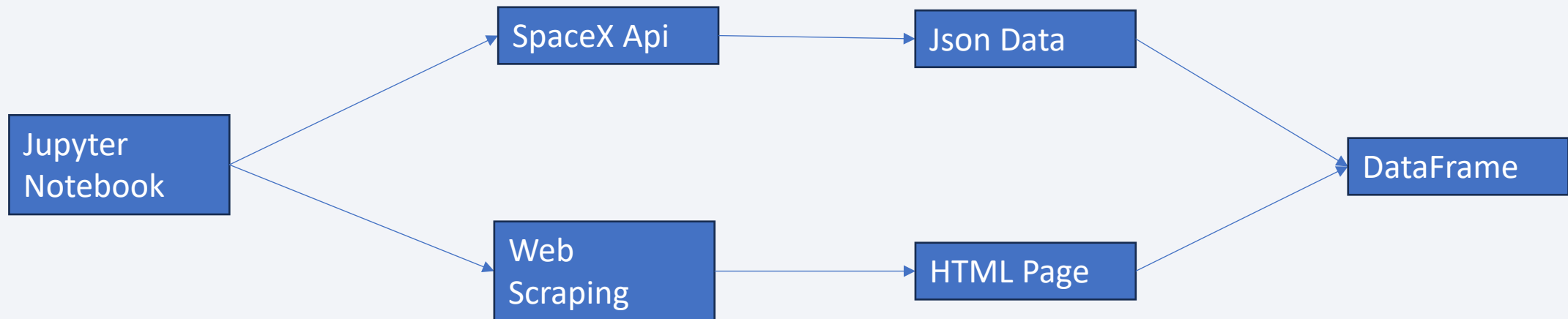
Executive Summary

- **Data collection methodology:**
 - Retrieve data from SpaceX Api and Web Scraping
- **Perform data wrangling**
 - Clean data for missing and incorrect data. Transform categorical data to numerical if necessary.
- **Perform exploratory data analysis (EDA) using visualization and SQL**
 - Calculate statistical properties for the collected data by using python modules and SQL. Gather better insights from the visualizations.
- **Perform interactive visual analytics using Folium and Plotly Dash**
 - Gather information about Launch site, geographical factors, vicinity and effect on success rate.
- **Perform predictive analysis using classification models**
 - Investigate various predictive models with optimal parameters and select best one for the future predictions.

Data Collection

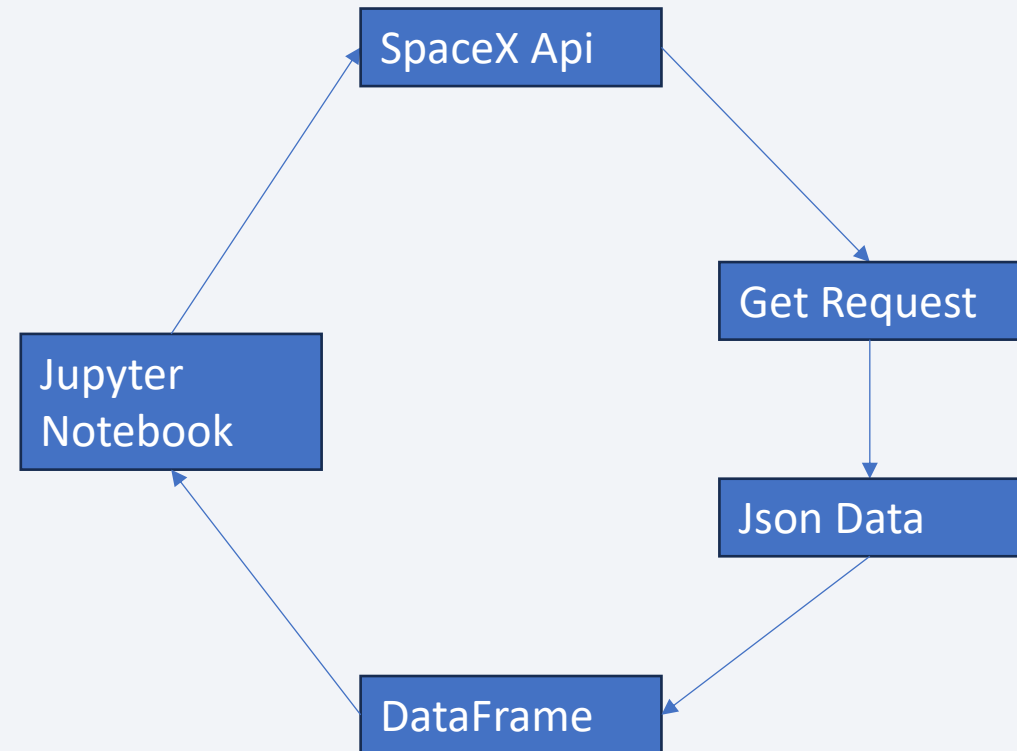
Data was collected by two methods:

- SpaceX API – past launch data and individual results were obtained from several SpaceX API endpoints.
- Wikipedia – web scraping on wiki page was used to get list of Falcon 9 Launches.



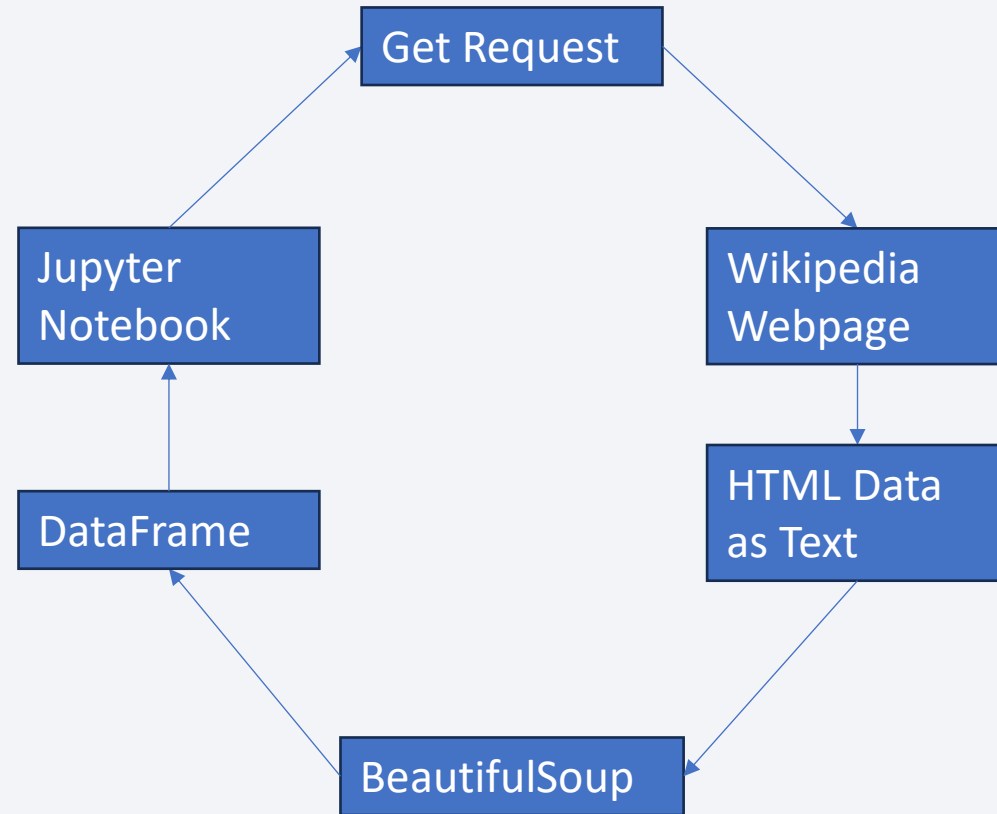
Data Collection – SpaceX API

- The json data is obtained from SpaceX Api using request library. The json is normalized and converted to pandas Dataframe for further investigation.
- Github link for the web scraping notebook: [SpaceX API Notebook](#)



Data Collection - Scraping

- The webpage from Wikipedia was received by HTTP get request and BeautifulSoup was used to parse and populate pandas DataFrame.
- GitHub URL of the completed [web scraping notebook](#)



Data Wrangling

- After data was loaded as DataFrame in Jupyter Notebook, following steps are used for data wrangling:
 - ✓ Identifying missing data. Remove or add data appropriately for missing data.
 - ✓ Identifying numerical and categorical data
 - ✓ Transforming categorical data to numerical data as required
 - ✓ Calculate various statistical parameters and gather insights.
- [Completed data wrangling notebook](#)

EDA with Data Visualization

- Various charts are used to represent the collected data to gain important insights. Charts are used to determine the relationship between various features and identify most significant features that affect the target variable.
- Following charts are used:
 - Scatter plots
 - Bar Charts
- [GitHub URL of your completed EDA with data visualization](#)

EDA with SQL

- Following SQL queries are used to obtain information from collected data:
 - Get unique launch sites
 - Show 5 launch sites that start with “CCA”
 - Calculate total payload mass for NASA
 - Calculate mean payload carried by F9
 - Get the date for first successful landing
 - Get the boosters with payload mass between 4000 and 6000
 - Get number of successful and failed landing outcomes
 - Show boosters that carried max payload mass
 - Get monthly list of landings for the year 2015
 - Get landing outcomes from 2010-06-04 to 2017-03-20
- [GitHub URL of your completed EDA with SQL](#)

Build an Interactive Map with Folium

- Markers, Circles and Lines are used to identify geographical patterns for different launches. The objective is to check the relation between the launch outcome and geographical conditions. It also helps to identify where maximum number of launches are performed.
- [GitHub URL of your completed interactive map with Folium map](#)

Build a Dashboard with Plotly Dash

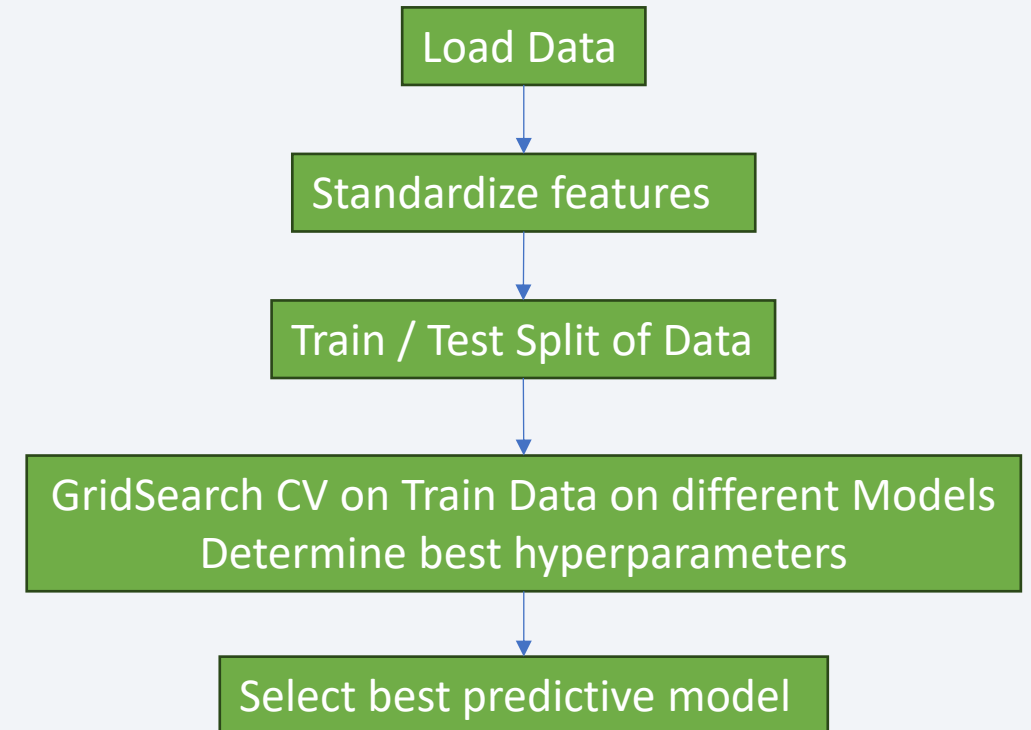
- Two plots are used in plotly dash Dashboard to represent the finding of the data:
 - A Pie Chart – It is used for visual representation for success and failure outcome for launches on all sites or specific sites.
 - A scatter Plot – It is used to show relation between launch site, payload mass and booster version.

A dropdown is used to select individual site or all sites. A slider is used to select payload mass range.

- [GitHub URL of completed Plotly Dash lab](#)

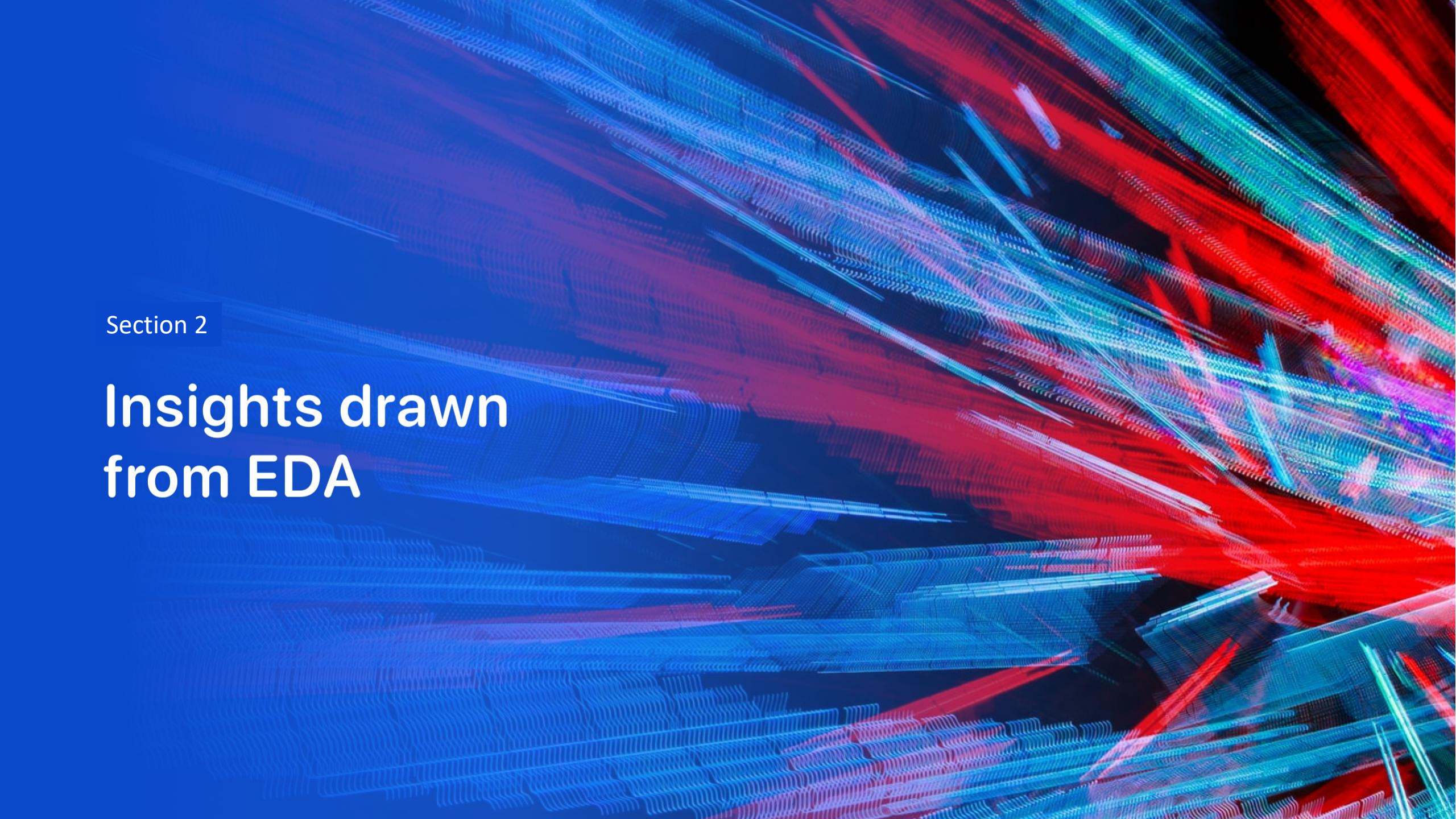
Predictive Analysis (Classification)

- The data loaded in DataFrame and features X is standardized by StandardScaler and target variable Y is converted to numpy array.
- The data is separated to train and test data.
- Following models are investigated and GridSearchCV is used to determine best fitting hyperparameters:
 - Logistic Regression
 - SVC
 - Decision Tress Classifier
 - KNN Classifier
- [GitHub URL of completed predictive analysis lab](#)



Results

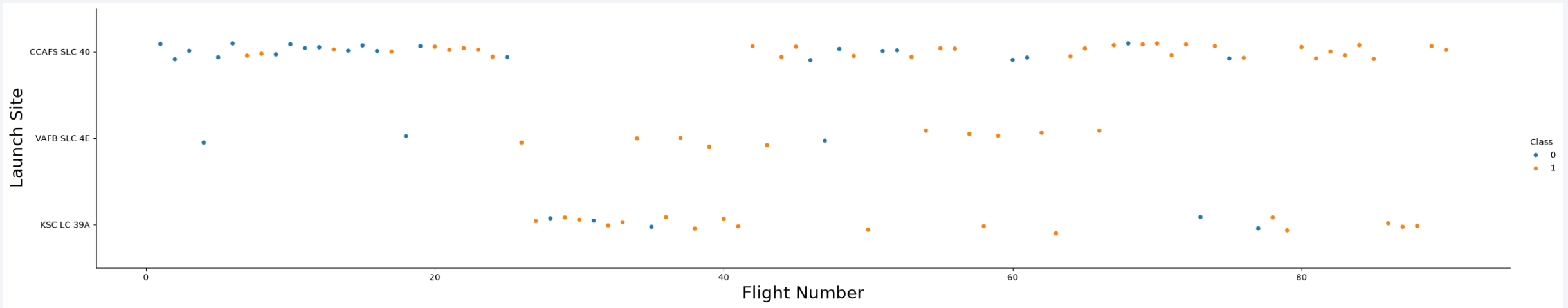
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

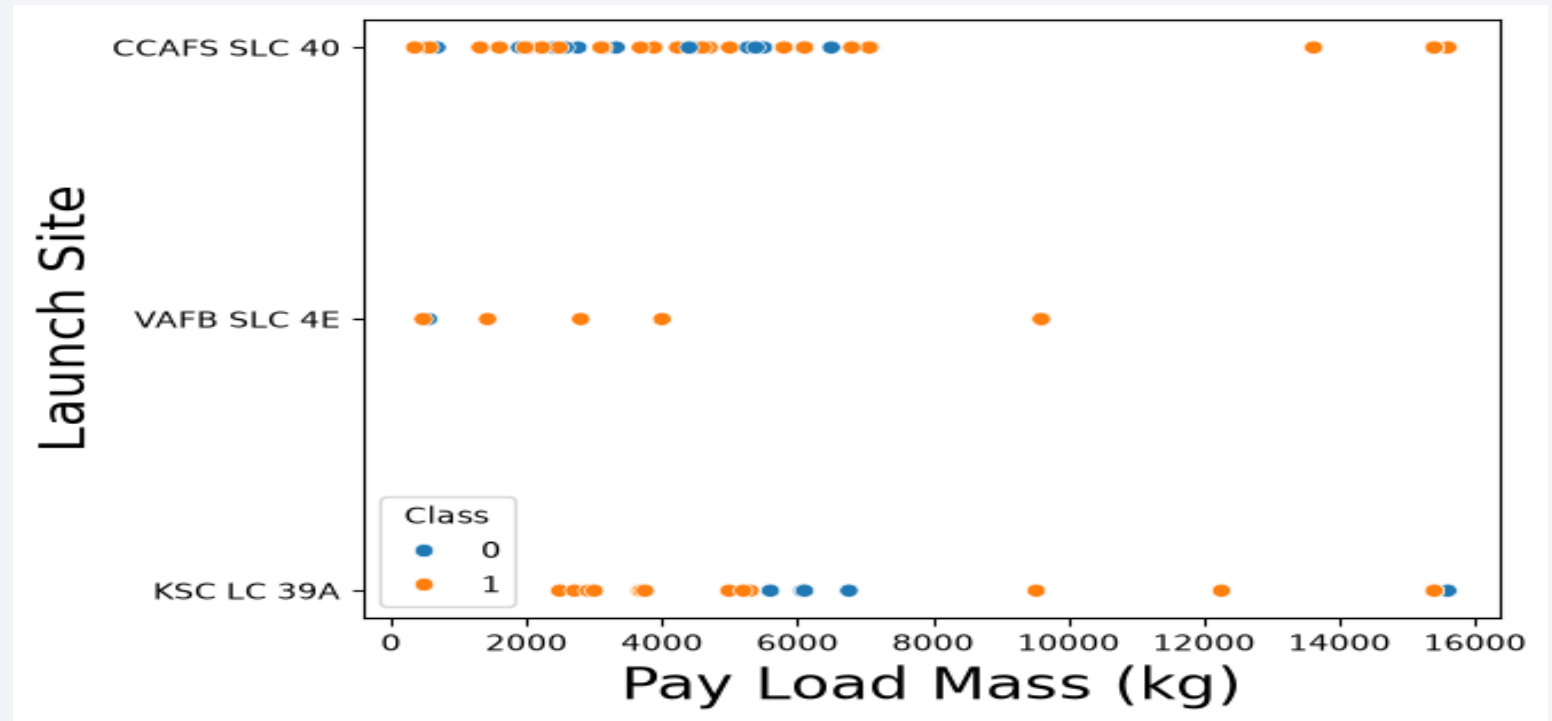
Flight Number vs. Launch Site



- Flight Number has no clear relation with success rate. However, flights after number 80 show 100% success.
- Flight number has no relation with Launch Site.
- However, total flights in site CCAFS SLC 40 is larger than other two sites.

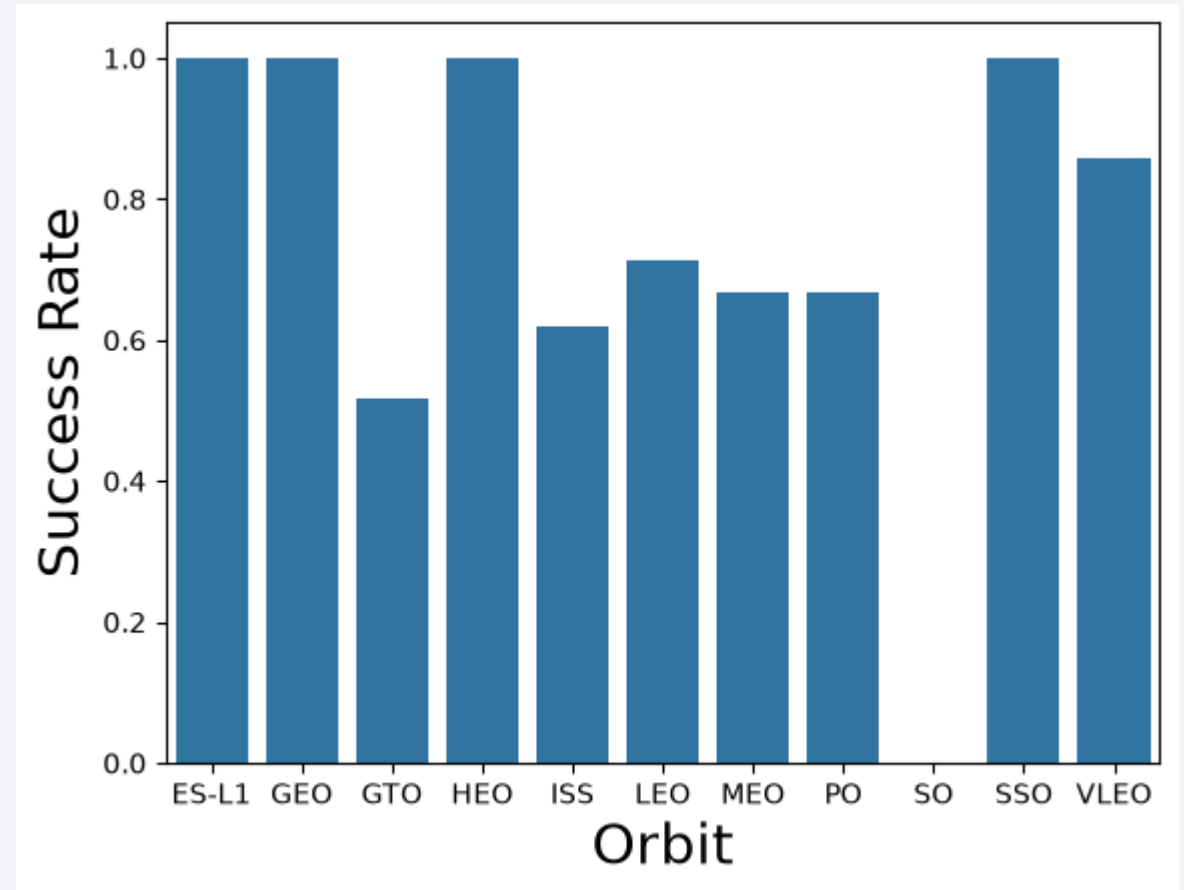
Payload vs. Launch Site

No clear relationship between payload and launch success rate for all sites.



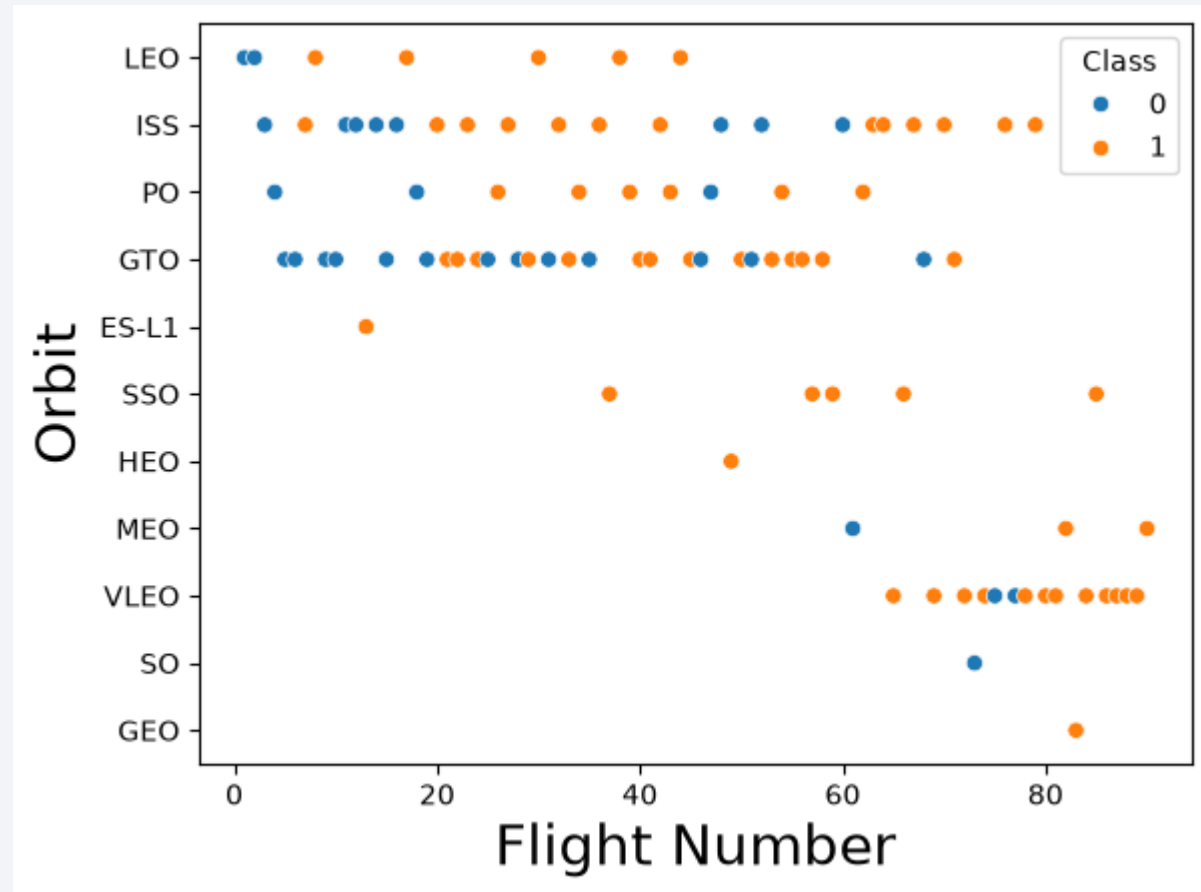
Success Rate vs. Orbit Type

- ES-L1, GEO, HEO and SSO orbits have high success rates.
- GTO has the least success outcome
- SO orbit has not enough data to be evaluated properly.



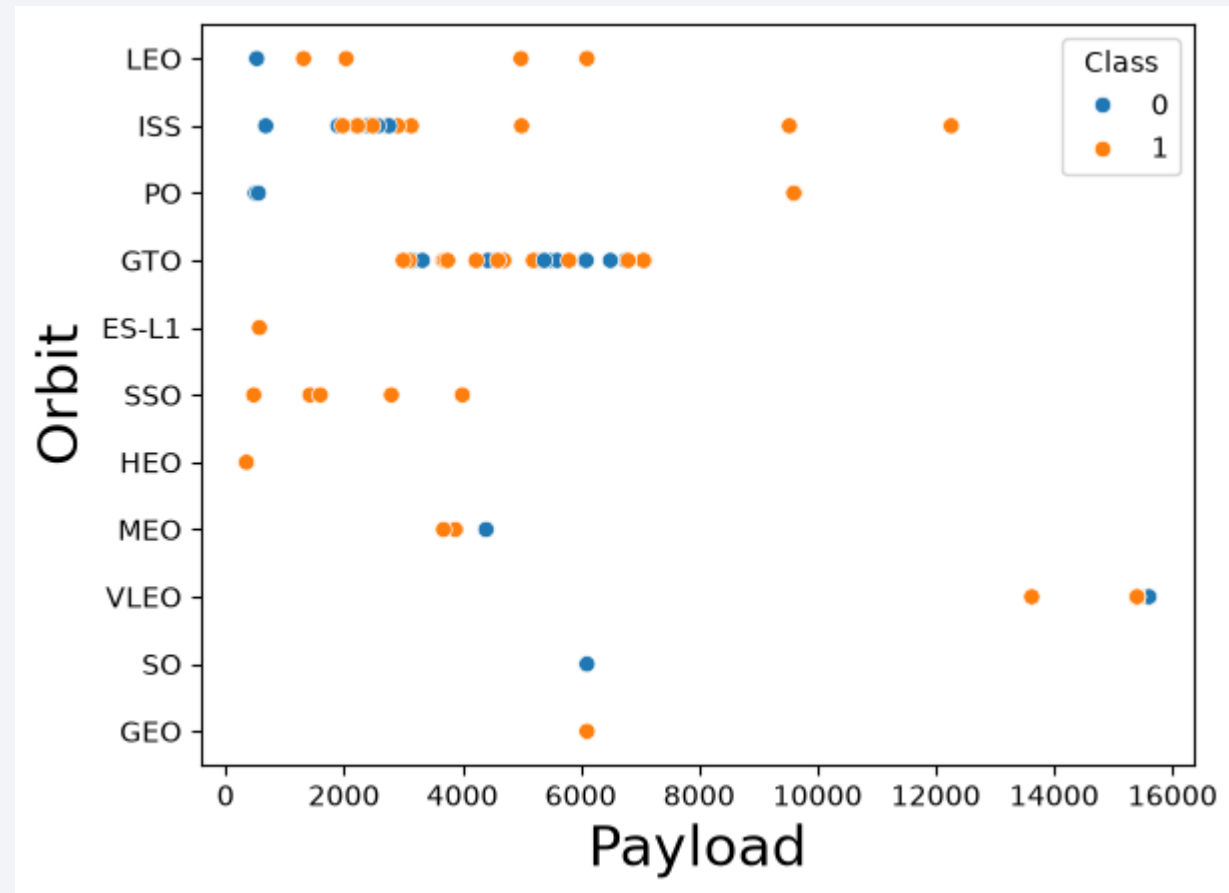
Flight Number vs. Orbit Type

- Flight numbers are scattered on various orbit types.
- Improved success rate for higher flight number indicate technological improvements and experience.



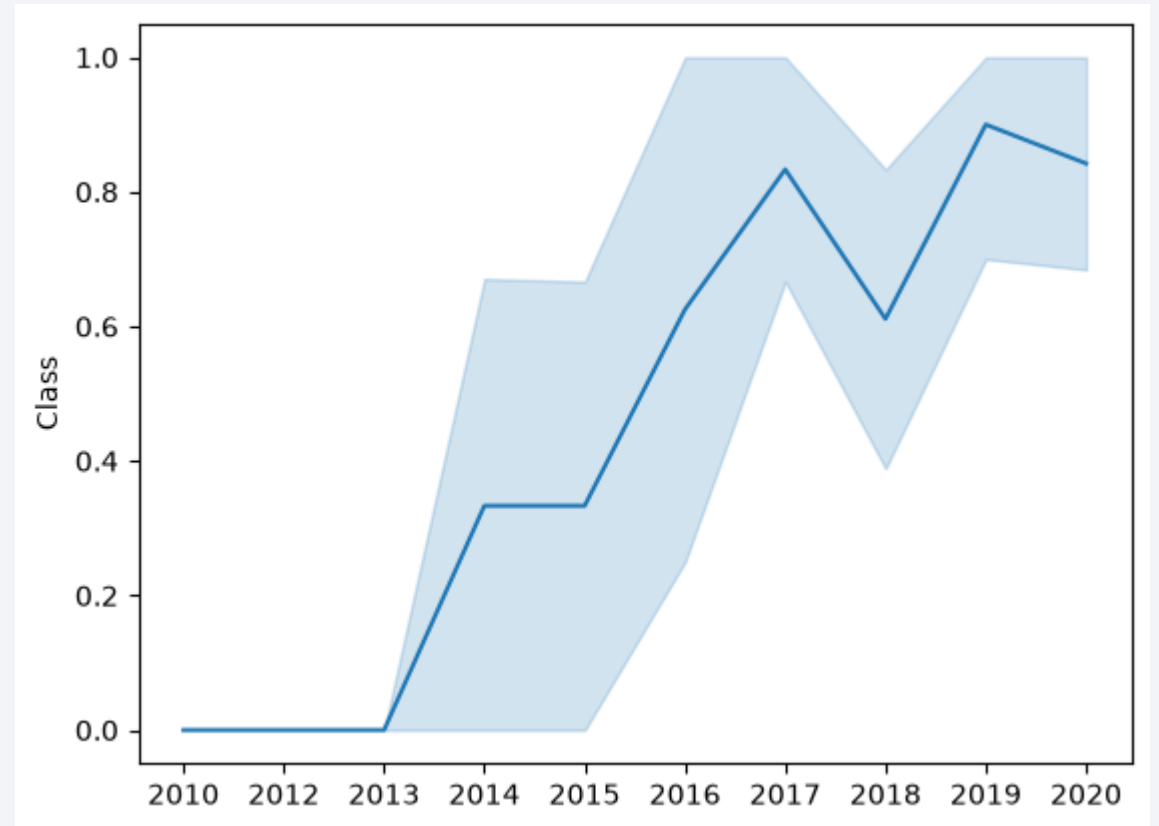
Payload vs. Orbit Type

- Payload does not show direct influence on success rate.



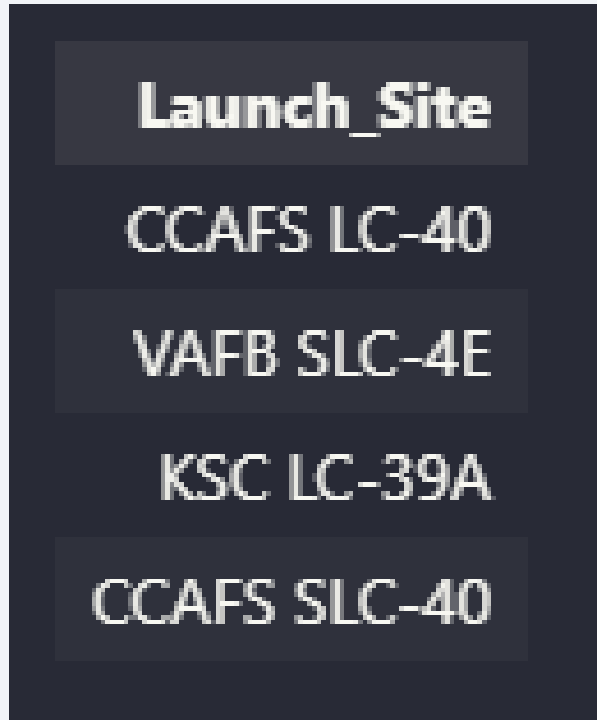
Launch Success Yearly Trend

- Success rate is increasing gradually over the years showing setback in year 2018.
- Increased success rate may be due to the lessons from past experience and technological advancements.



All Launch Site Names

- There are four sites for rocket launches as shown in this picture.



Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- 5 records where launch sites begin with `CCA`

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Total payload mass for NASA (CRS) is 45596 kg.

```
sum(PAYLOAD_MASS_KG_)
```

```
45596
```

Average Payload Mass by F9 v1.1

- The average payload mass carried by booster version F9 v1.1 is 2534.7 kg.

```
avg(PAYLOAD_MASS_KG_)
```

```
2534.66666666666665
```

First Successful Ground Landing Date

- The date of the first successful landing outcome on ground pad is 2015 December 22.

min(Date)

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- Four boosters have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 kg.

Booster_Version	PAYLOAD_MASS_KG_
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

Total Number of Successful and Failure Mission Outcomes

- The total number of successful outcome is 100 and failure mission outcome is 1.

Mission_Outcome	count(Mlssion_Outcome)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- The booster that carried maximum payload are show in the picture below:

Booster_Version	PAYLOAD_MASS_KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

- Two failed landing outcomes in drone ship for year 2015 are in January and April.

Month	Landing_Outcome	Booster_Version	Launch_Site	Date
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40	2015-01-10
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40	2015-04-14

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order is shown below. No attempts are made for maximum numbers.

Landing_Outcome	counts
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

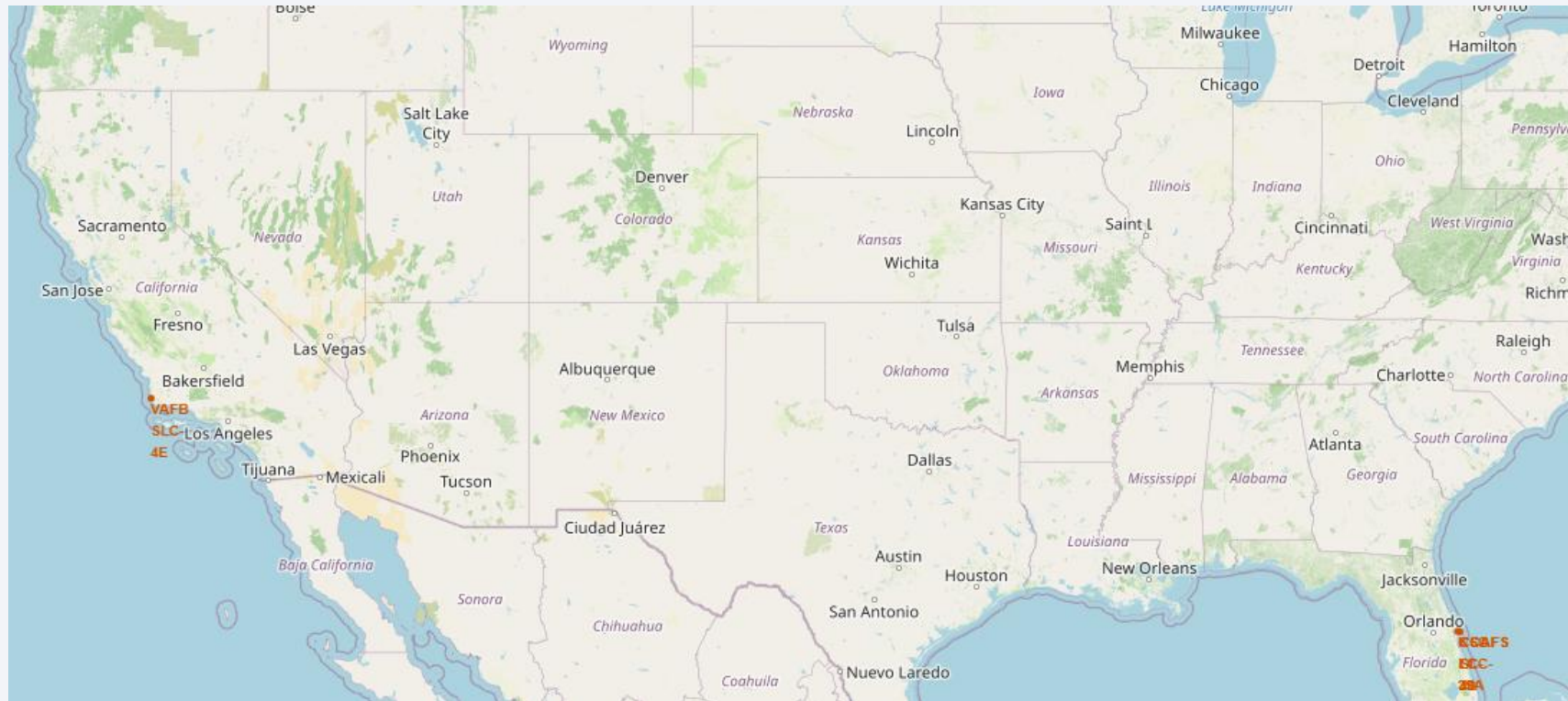
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left shows a clear blue sky.

Section 3

Launch Sites Proximities Analysis

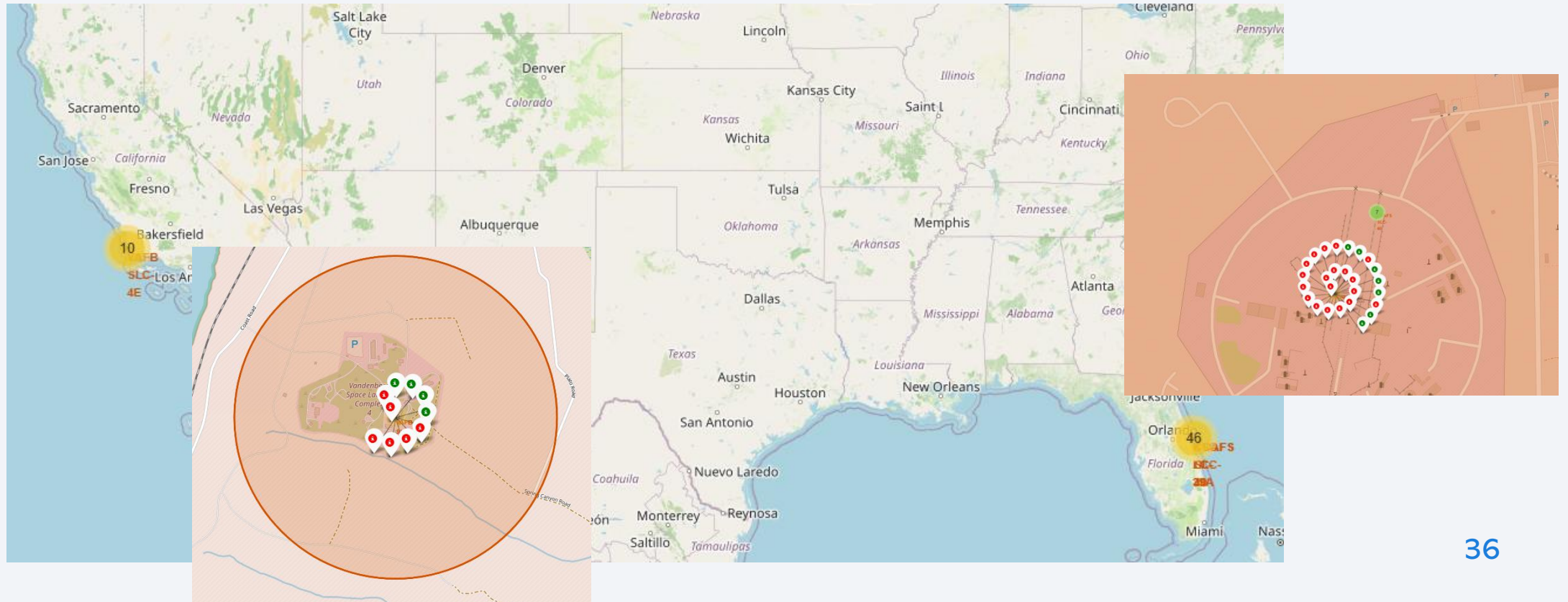
Launch Site Locations

- Launch sites are located near coastal regions with sufficient infrastructure.



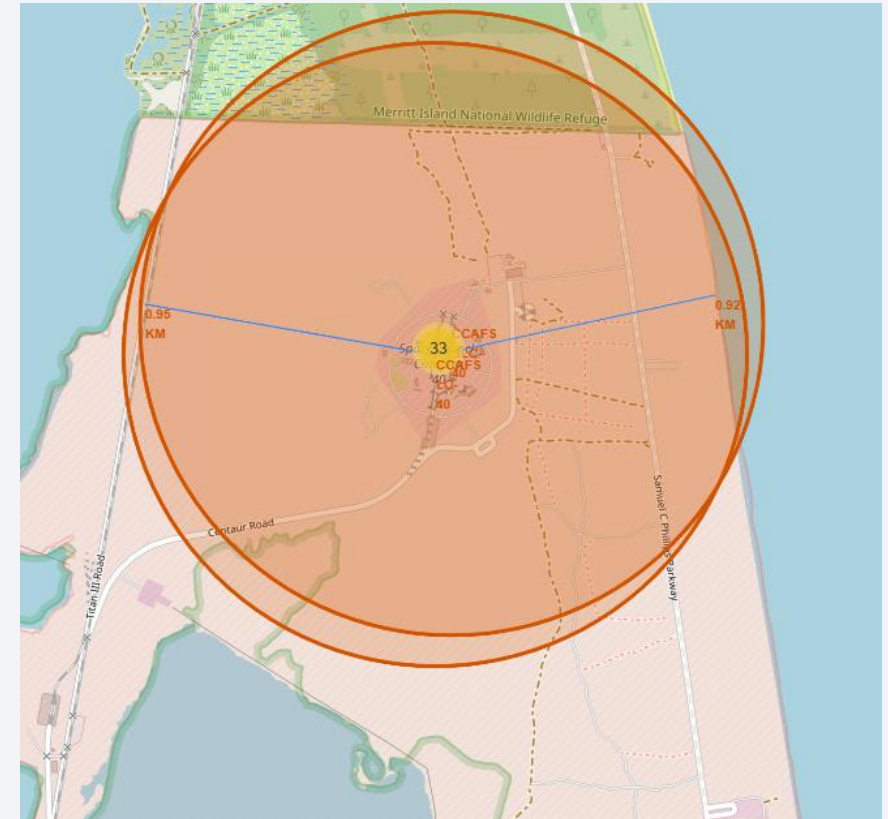
Launch Outcomes

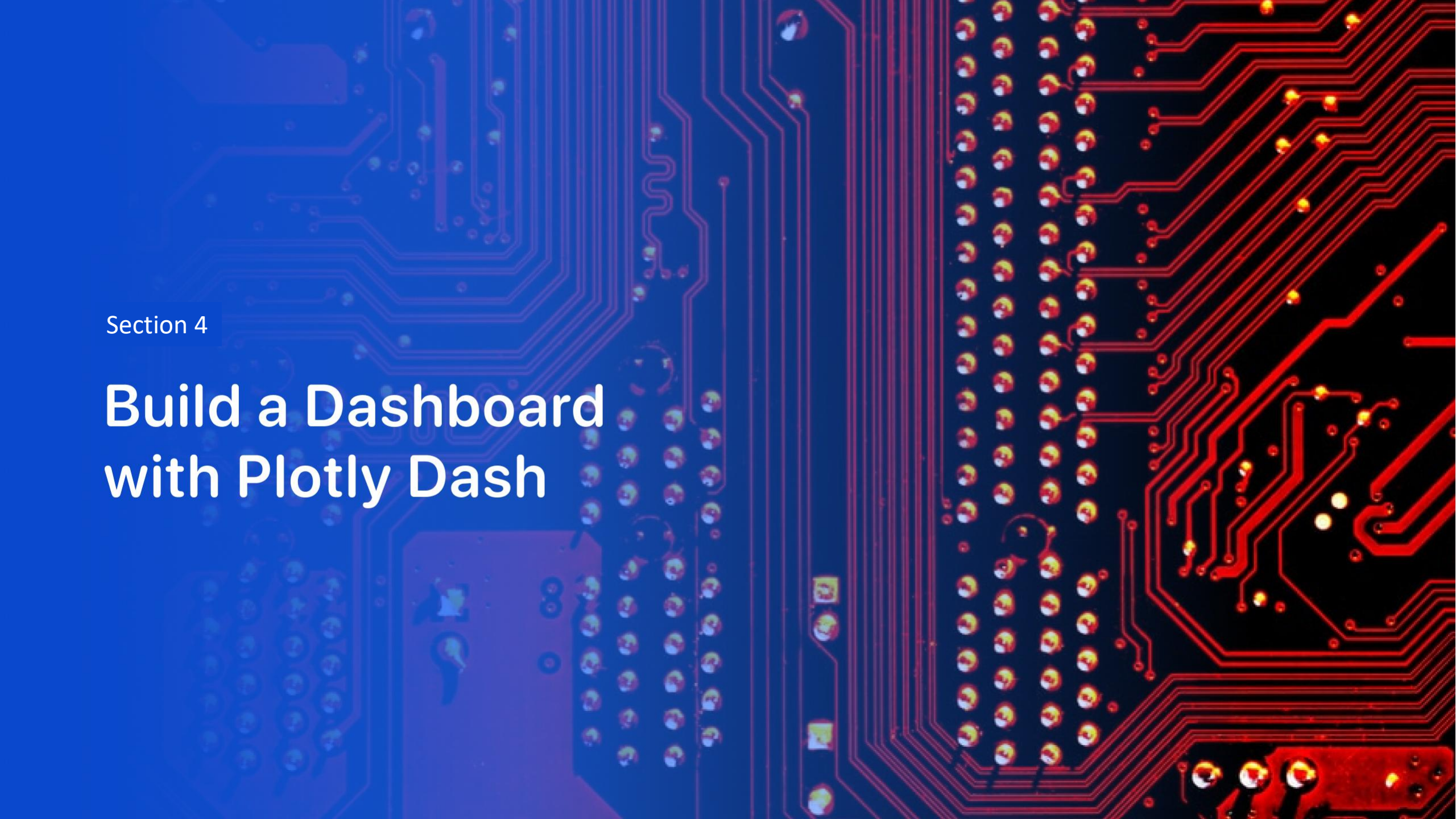
- Launch outcomes for the sites are shown in following pictures. Green markers indicate success and red marker indicate failure.



Proximities

- The distance to the major infrastructures can be calculated as follows. The launch sites are located near coastal regions and infrastructures like railway lines.



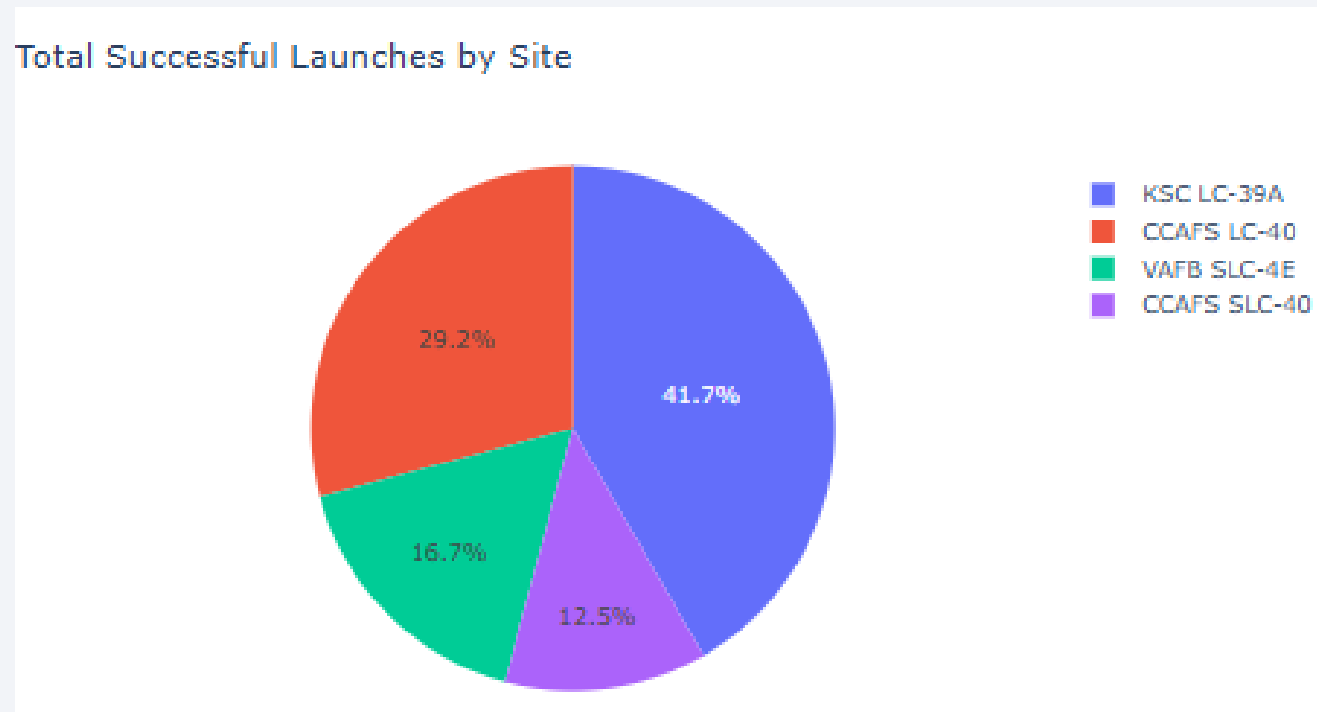


Section 4

Build a Dashboard with Plotly Dash

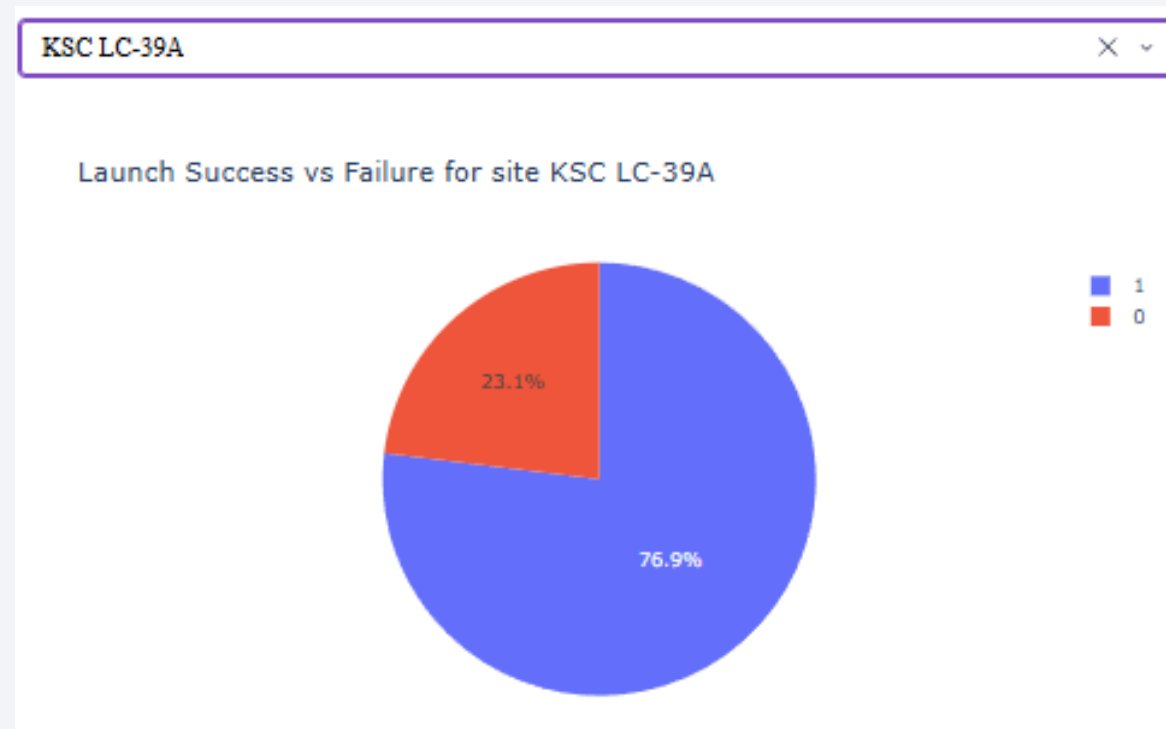
Successful Landings for Launch Sites

- KSC LC-39A has most successful landings and CCAFS SLC-40 as the least.



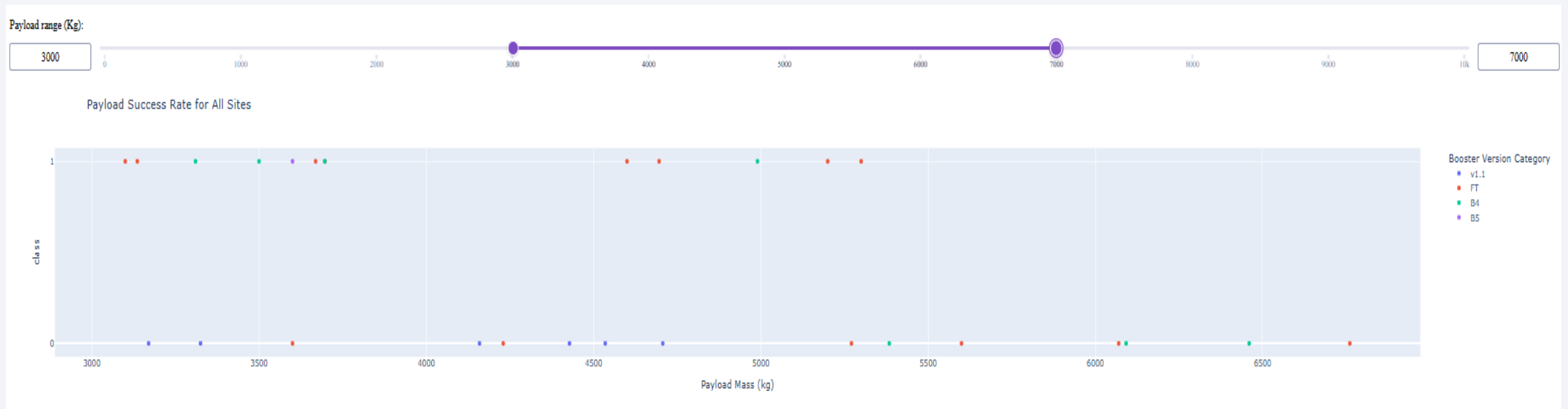
Launch Success – KSC LC-39A

- Launch site KSC LC-39A has 76.9% success rate.



Payload vs Success Rate

- The plot shows relation between payload mass and success rate for all sites for the range of 3000-7000 for payload mass.

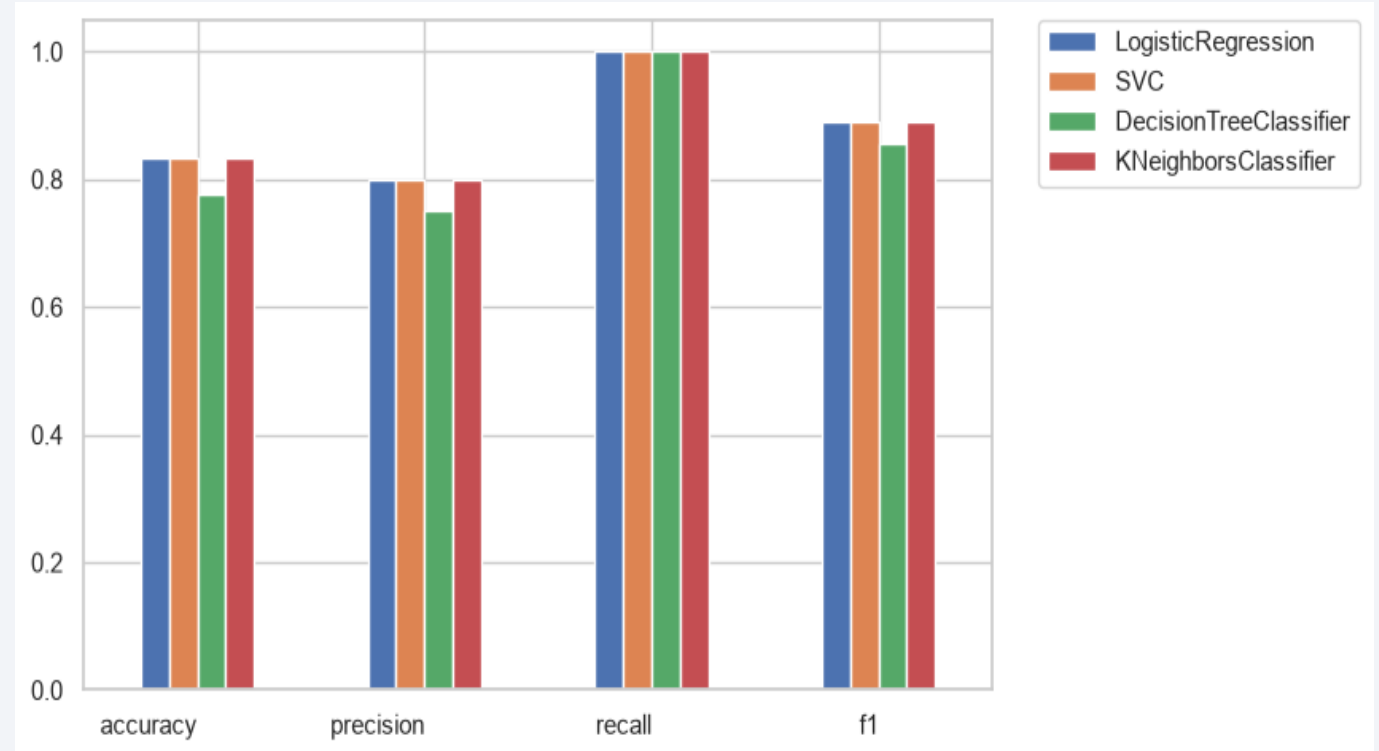


Section 5

Predictive Analysis (Classification)

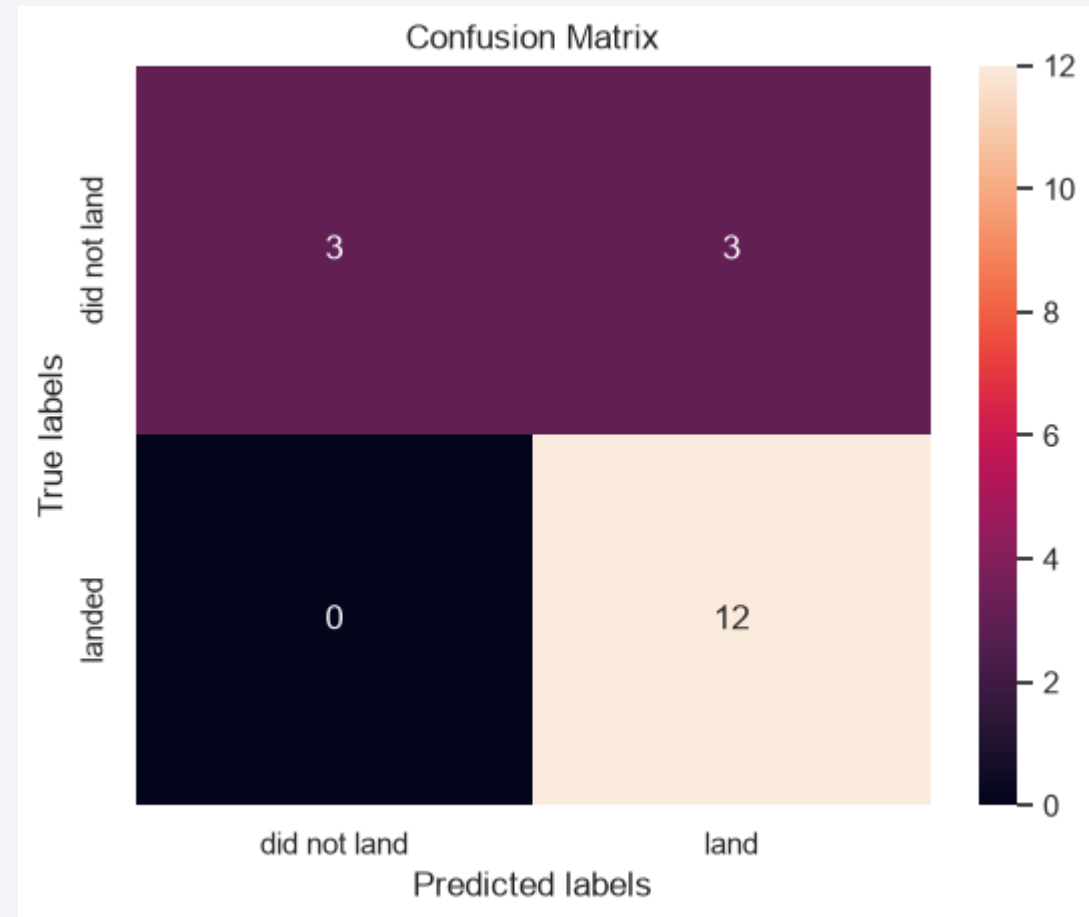
Classification Accuracy

- The plots shows similar accuracy scores for Logistic Regression, SVC and KNN Classifier.
- Decision Tree Classifier has lower accuracy and precision score for current data relative to other models.



Confusion Matrix

- 12 observations were predicted as true positive.
- 3 observations were predicted true negative and false negative each.
- No observations were predicted as true negative.



Conclusions

- Success rate shows improvement over time indicating progressive development and technological advancement.
- Success rate shows direct relation to orbit type and booster version.
- Launch site affects the success rate to some factor. KSC LC-39A performing the best.
- Among several predictive models, Logistic regression is selected with best results for accuracy, precision, recall and f1 score and its simplicity.

Appendix

- [Space X API](#)
- [List of Falcon 9 and Falcon Heavy launches](#)
- [CSV for Dataset after Data Wrangling](#)

Thank you!

